

Lecture Homework Week 06 - Thursday Template

Student Information

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Student Responses

Answer the following questions

Question 1

When running the Keyboard Demo in QEMU after introducing the bug, briefly describe how it behaves differently than before. Try a few keystrokes in the QEMU window and describe what you see :

The characters, while input, do not display in sequential lines, rather they overlap on the same line. Additionally, there are small graphical glitches throughout the screen, where letters pop in and out of the display. When I press enter to output the full character, there is no output.

Question 2

What do you think is happening? The program continues to execute, but what happens after it returns from `irq_handler`? What C subroutine gets called over and over?

The value in `pc` is set to what should be the value in `r5`, but since we did not pop `r4-11`, `pc` gets an incorrect value. This means that we will repeatedly return to random addresses. There is no C subroutine that we repeatedly call back to.

Question 3

Look at the terminal where the buggy version is running, what messages are being printed out about the timers? Do the timers work in this buggy version?

There are no messages being printed out about the timers, other than the `timer_init()` function beginning, and `timer_start 0` with a base address, meaning the timer does not work.

